

# AUTOMATIC FALL DETECTOR (GYRO)

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## ABSTRACT

To provide for the elderly to be rescued from the fall because the elderly died from the fall. It is second only to traffic accidents. Because they did not receive immediate assistance This thesis has developed a fall detection device. That is worn around the torso of the elderly When abnormal acceleration is detected, an alert will be sent through the system. Wi-Fi to Cloud Networking And the message will be sent to the Administrator's Line Application where the administrator can send messages from Line Application to the fall detection device. Likewise, the designed device uses the ESP8266 as its main processor and uses the MPU 6050 accelerometer module to detect a fall using the principle of free fall (Freefall) has been applied in the testing of the inspection. Falling Create a device to simulate the falling of the elderly by comparing the results from the experiment. Physics calculation results to confirm the ability to detect falls and it is used to adjust the parameters of the equipment for the most accurate fall detection.

## 1. INTRODUCTION

Currently, the Wireless Sensor Network is a technology to More popular communicators. The use of tiny sensor devices Many to measure properties such as temperature, heat, relative humidity of the environment that we are interested in and the results of information They come to create new knowledge. In order to automatically respond to changes in the environment. This wireless technology can be applied in many fields including agriculture. Industrial Security And in medicine. In medicine, wireless sensor networks are used in many applications, such as monitoring patients via wireless networks. This system uses sensors that can measure the values needed for treatment such as Heart rate values to assist in the examination and treatment of patients. And another side that has the application of 2 wireless sensors is the use of the elderly. The application for wireless sensors and analysis. The movement characteristics of the elderly, how the elderly behaved at that time, such as standing, running, walking, lying, sitting or falling down. Which if we can detect these behaviors, it is particularly useful for. Reduce accidents that may occur with the elderly or with notification if an accident occurred in the case at that time. There is no elderly caregiver.

Development of hardware to detect fall behavior using 3-axis sensors by measuring the accelerometer in the x, y, and z axis.

Analysis of apartheid movement of human behavior. The acceleration values in the x, y, and z axes are analyzed through the Simulink block using the Threshold that has been tested to detect the behavior of falls. Which is the principle that It is appropriate to analyze the data of the human behavioral behavior that needs to be quickly reported.

## 2. RELATED WORK

From the statistics of the falling of the elderly as mentioned above to do Have a group of researchers to focus on and present research work related to the detection of elderly falls. It is presented as a research article in both domestic and international journals. Which can summarize the pattern of fall detection into 2 types, which are Wearable devices that must be next to the operator and devices not to be attached to the user.

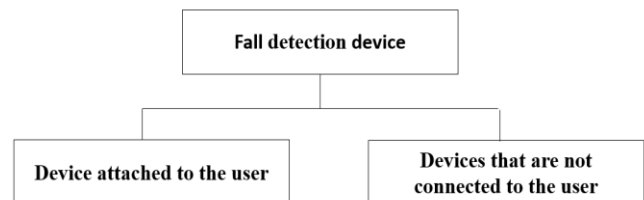


Figure 2.1 Classify the fall detection device

Elman Neural Network Identify Elders Fall Signal base on Second-order Train Method This research is to detect falls by leading Offers an Elman Neural Network method by receiving the input signal from the accelerometer sensor attached to the sensor, try out this simple method based on regressive learning with the input signal being taken at the present time. The moment and the previous input signal (which occurs before) are used to improve neural network processing. In this research, we have designed to Allow the device to take the subject and a notification via Bluetooth (IEEE 802.11) signal to a personal computer When the opponent is put into the device.

This chapter of the project discusses the nature of the falls, including the risk factors for Fallen in the elderly to guide the design of the fall detection device, the developer has taken into account the

appearance of the device that should be compact. Function to use at Ease of use and the convenience of use when the user wears the device. The target audience of using this fall detection device is the elderly group who lived alone without caretakers.

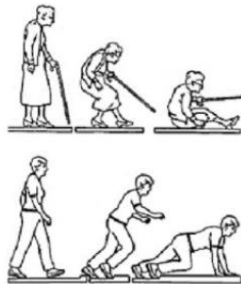
## 2.1 Falls in the elderly and the risk factors that caused the crash

### 2.1.1 Falls in the elderly

A fall kills the elderly as the second highest in accidental injuries [1] after road accidents. Falls include falling on the ground level, slipping, stumbling, stumbling, being knocked down by others. Or falls at different levels, such as falling stairs, falling a chair, etc., data from the World Health Organization [2] indicates that the elderly aged 65 and over are 28-35 percent more likely to fall annually and increase to 32-42 percent. When you step into the 70th year, you will notice that if you increase your age, your risk will be High in falls Especially those who have symptoms of muscle weakness Balance problems: Falling can cause hip or hip fracture to occur. The main spine can be fractured in people with osteoporosis. If the head hits Basically, it can cause cerebral hemorrhage, which can lead to disability and subsequent death.

### 2.1.2 Nature of fall

The nature of falling depends on the age, shown in figure 2.2 during adolescence or working age group. Able to walk fluently Will have a feature of falling forward with your hands to lean the ground Causing the risk of A fracture in the forearm the elderly is agile, slow to walk, often lose their balance, fall backward, resulting in hip fractures. And pelvis Which will result in Injury is severe and can even be fatal if the elderly does not receive immediate assistance.



**Figure 2.2** Fall that occurred in the elderly (top picture) and working age people (bottom picture)

### 2.1.3 Risk factors causing falling

Falling can be caused by a variety of causes. If the type of risk factor is categorized into two categories: internal and external. Internal factors are factors that arise from the elderly themselves in the health and behavior of the individual. For example

1. Inadequate muscle strength
2. High blood pressure or lower than the standard.

3. Problems with urinary incontinence

4. The ability to see has decreased.

5. Abnormal balance such as unequal water in the ears  
Cerebrovascular patients

6. Side effects from drug use such as sleeping pills, drugs for depression patients.

The external factor is the environment outside the elderly. Environment within the address If the residents are not favorable, it will result in an accident that contributes to the increased risk of falls.

Therefore, we should prevent falls in the elderly as follows.

1. Folk areas Should take care of the floor to dry, not use oil and polish. Objects should not be placed in the pathway, in areas with a level floor, they should be clearly marked.

2. The bedroom is tidy. The bed should be at a level from the floor. Suitable to make it easier to get out of bed

3. Toilets, install handrails in the toilet or shower area. The bathroom door should be locked at Can be opened from the outside in case of an emergency, they will be able to help immediately.

4. The kitchen keeps the things that are needed in a convenient rack. Heavy objects should be placed on the floor without obstructing walkways.

5. In the stairwell area, avoid going up and down the stairs unless necessary. If necessary, Choose a durable staircase. Has enough illumination and there should be no obstructions

6. Other factors that are factors such as dress should wear appropriate clothing, clothes that fit the body. Wear to aid in the comfort of movement Mobility aids, such as canes, wheelchairs, should be strong and able to support the weight of the elderly. It should also be practiced and properly used. This is done regularly to gain expertise and familiarity in the use of the equipment.

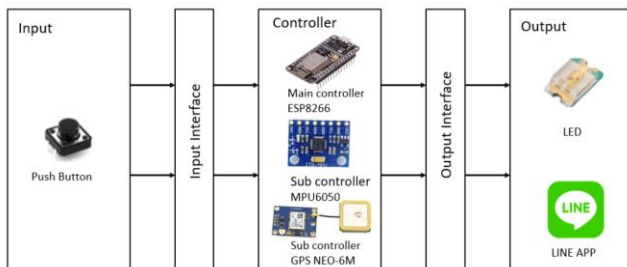
In the past, a group of researchers has developed a widely used fall detection device, the fall detection device presented in this thesis is classified as a user-adjacent fall detection device. User privacy and ease of use Device work in research involving this type of fall detection device, researchers have focused on a fall detection that uses high performance computing resources. Because it uses neural network processing [3], the device is not bulky. Suitable for portability It also has a very low usage time per battery charge. Subsequently relevant research, the research group has focused on reducing the number of wearable devices. Wearing or using it with the elderly, smartphones used in daily life have been used to detect falls [4]. However, if the elderly does not carry their smartphones all the time. In addition, each smartphone has different power processing and acceleration sensor limitations. This research and the

last order of relevant research is to develop a fall detection device that Two sensors are used for detection [5] by this design, there are limitations on the user side. Lack of flexibility for daily life because there are many devices attached to the user.

Therefore, in the design of the fall detectors presented in this project, the developers have designed a single sensor to increase user dexterity as well as reduce the consumption of computing resources. Because the device is a wearable device requires low battery life. The most to increase the service life with an overview of the design as follows.

## 2.2 Design overview of the proposed fall detectors

The elderly fall detection device presented in this project was designed under the concept of “The Elderly Fall Detection”. Operators must wear comfortable and easy to use. Therefore, it focuses on designing the size of the device. It has a compact size suitable for wearing. Supports 2.4 GHz Wi-Fi connection (IEEE 802.11 b / g / n), it can detect a fall immediately and can let the user press the button for help. It is sending a message through the Cloud Network, whereby a request for assistance messages will be sent to the Line Application on the caregiver's smartphone in order to let the caregiver be aware of the incident and assist in a timely manner. Detection device design Fall has chosen to use the equipment as shown in Figure 2.3.



**Figure 2.3** Details of components inside a proposed fall detector

Proposed fall detection device design so that the design complies with the proposed fall detectors. The design that has been mentioned above One push-button switch has been selected to perform various functions and request assistance. ESP8266 is used as a common processor with the MPU6050 accelerometer. User interaction is also available. LED is used to display optical signals. Can divide the proposed fall detector components can be divided into five parts

1. The design of the input section consists of a push-button switch that accepts user commands and allows the user to press the switch for help and to select functions in the device settings.

2. The processor design consists of an ESP8266 used to process the fall detection sequence. The work order is the state. The MPU6050 module receives commands from the input section who is wearing the device.

3. The design of the output part consists of LED; it is used for signaling to inform the elderly. And the caregiver is aware of the operating status of the device by using light from the LED.

4. Design of the power section A Lithium polymer (Li-Po) battery has been selected as a power source for the device.

5. Printing board design and the case of the device Obtained in the design of the print sheet section Has chosen to use the design of the case of the device is designed to be rounded and small, suitable for portability. Input design.

Concept of designing inputs to receive commands from users. Has taken into account the format Simple operation of the device due to the elderly users Therefore has designed the device to Accepts input from the pushbutton (Push button) from the user's push of only 1 button, which is assigned to these pushbutton Press commands to perform the following functions. Default device settings Accept and cancel from notification when a fall occurs and seek help when there is an emergency Processor design.

The processor design takes into account the features of devices that can be connected via Wi-Fi in the 2.4 GHz band to provide an alert for different device statuses, including notifications. In case of emergency in this thesis, the ESP8266 module was selected as the main processor because the module has a Wi-Fi connection circuit built into the module, thus eliminating the need for external circuits. The fall detection section uses the accelerometer module MPU6050 to detect the state of the fall. If considering the acceleration of a severe fall, it can be compared. The body of the falling person is like an object that has free fall with the g-force of the earth with a period of time. Of the fall varies with the height of the falling in which to measure this kind of fall within the module.

The MPU6050 has an internal processor that can perform an independent fall detection function. Once the MPU6050 module detects an independent fall situation (fall), it sends an interrupt signal to the ESP8266 module to inform the state that the air has occurred. Immediately, when the ESP8266 module receives an interrupt signal, the failure warning will send a message for help via the group network. Continue to moderators.



**Figure 2.4** shows the design of the device case

### 3. MATERIAL AND METHOD

Within a room or in a user-occupied area, there are fall detection systems, from the device alarm, using the gyro sensor to measure the change of the device. This system concept that is shown in Figure 3.1 can be accomplished through the following stages.

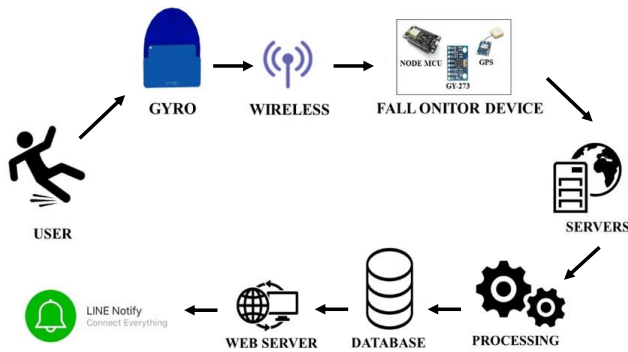


Figure 3.1 The new system in project

#### 3.1 Use case diagram

The caretaker must be equipped with the device and being monitored while the caregiver can monitor the caretaker and medical experts can get the hotline.

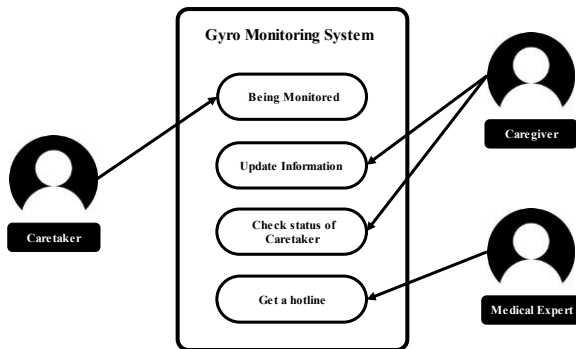


Figure 3.2 Use case diagram

#### 3.2 Flowchart

NodeMCU repetitively collects data from MPU 6050. NodeMCU does calculations to check if orientation data reached a predefined threshold. If the threshold was reached, NodeMCU waits 10 seconds to delay if the patient is able to help himself and cancel the alert. After 10 second passed and fall detected had not been canceled, NodeMCU proceeded with the alert sending procedure.

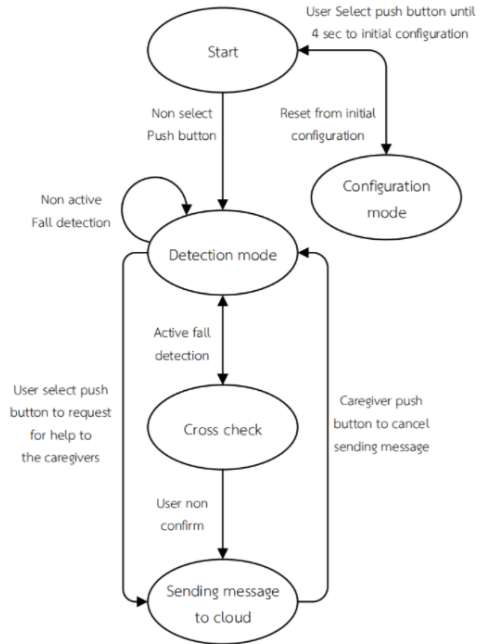


Figure 3.3 System's flowchart

#### 3.3 Circuit Diagram

The electronics components used in this project. The accelerometer (MPU 6050) is connected to NodeMCU. The node sends real-time information to the web server when fall is detected.

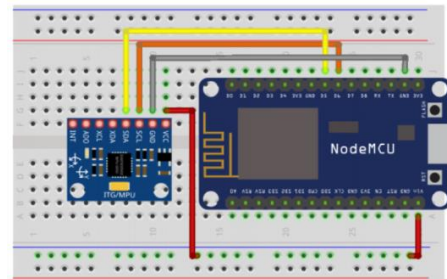


Figure 3.4 Circuit Diagram

#### 3.4 Database and Web application

##### 3.4.1 Data Dictionary

This table shows a database of both field name of caregiver and field name of caretaker.

Table 3.1 Caregiver and Caretaker's dictionary

| Field name | Data type | Field size | Description                                        |
|------------|-----------|------------|----------------------------------------------------|
| id         | int       | 11         | Identification number for information of caregiver |

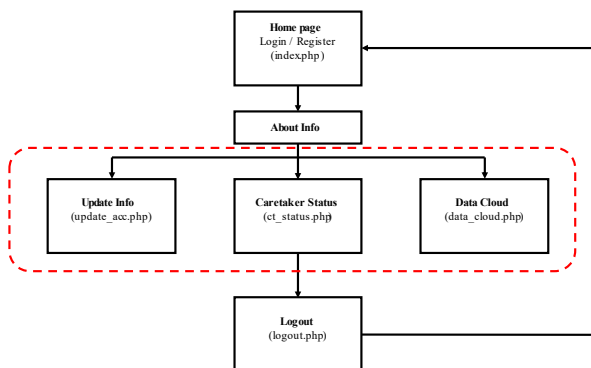
|           |         |     |                                                          |
|-----------|---------|-----|----------------------------------------------------------|
| user_name | varchar | 150 | Identification name for information of caregiver         |
| email     | varchar | 150 | Email of caregiver                                       |
| password  | varchar | 150 | Password that users use to login stored in hash function |
| fname     | varchar | 150 | First name of caregiver                                  |
| lname     | varchar | 150 | Last name of caregiver                                   |
| address   | varchar | 300 | Identification address of caregiver                      |
| phone     | varchar | 50  | Phone number contact of caregiver                        |
| fnamect   | varchar | 150 | First name of caretaker                                  |
| lnamect   | varchar | 150 | Last name of caretaker                                   |
| medinfo   | varchar | 300 | Medical information of caretaker                         |
| addressct | varchar | 300 | Identification address of caretaker                      |
| phonect   | varchar | 50  | Phone number contact of caretaker                        |

**Table 3.2** Sensor's data dictionary

| Field name   | Data type | Field size | Description                               |
|--------------|-----------|------------|-------------------------------------------|
| id           | int       | 11         | The number of times to collect the value. |
| value1       | int       | 10         | The value axis X                          |
| value2       | int       | 10         | The value axis Y                          |
| value3       | int       | 10         | The value axis Z                          |
| status       | varchar   | 50         | Fall state                                |
| value4       | float     | -          | The value of latitude                     |
| value5       | float     | -          | The value of longitude                    |
| Loc_NAME     | text      | -          | Place name                                |
| reading_time | timestamp | -          | Current retention date and time           |

### 3.6.2 Flow of Web Pages

A flowchart of a web application of web pages is shown in Figure 3.5



**Figure 3.5** Flowchart of web pages

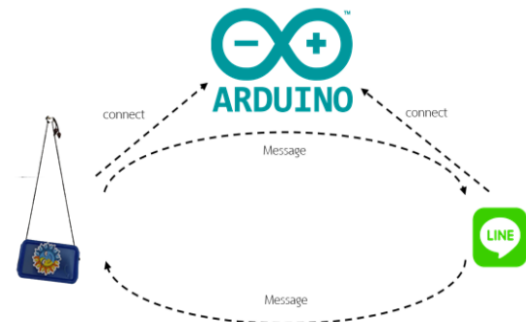
- The home page consists of login and register.

- If a caregiver logs into the system, the web page with a list of the About Info sub-menu consists of updates about caregiver and caretaker information, a map, and a graph of caretaker status.
- After log-out, the system will return to the home page.

### 3.7 Line Application

The detection device in this project can be considered. It is a device that is classified as an IOT device, Internet of thing, that is, the designed fall detection device can connect and transmit the fall alarm information via the Internet network. Another way to use the user can bring the device to use anywhere. Under the condition of connection of the device to be connected to a Wi-Fi communication network in the 2.4 GHz band(IEEE 802.11 b / g / n).

Manage the process of monitoring the health of the device, notifications, as well as account management. In this project, cloud service was used. In the process of sending notification messages, the service of Line Bot was used in conjunction with Arduino and ESP8266 in the process of sending messages from the cloud network to the device. Part of the process of managing automatic message responses and providing Web application services.



**Figure 3.6** Send and receive data within the ARDUINO

## 4. EXPERIMENTAL RESULT

### 4.1 Falling Detection Device

This project presents fall detection for the elderly by using accelerometer sensors installed on the torso around the user's chest, shown in Figure 4.1. The experiments were performed to find the optimal values for detecting various types of falls, such as sitting, standing, lying and bending.

Collected data obtained from various movements were analyzed to determine the acceleration that would cause the crash. In the analysis, the maximum value of the acceleration in each action was used to determine the initial acceleration to detect falls.



Figure 4.1 Design of the IoT Fall Detection Device

## 4.2 Web Application

We develop a website that supports our users to update their personal information, view caretaker status and view caretaker addresses. For this website, you need to log in to the system to use.

### 4.2.1 Home page

In the Home page, as shown in Figure 4.2, it shows our project name, contact, login, and register.

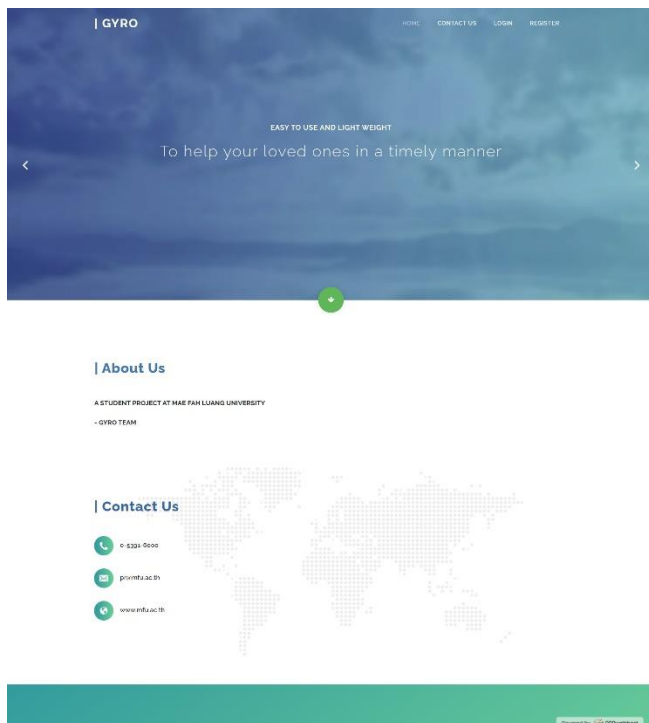


Figure 4.2 Home page

### 4.2.2 Register page and Login page

We have a different page for register and login as shown in figure 4.3 and 4.4.

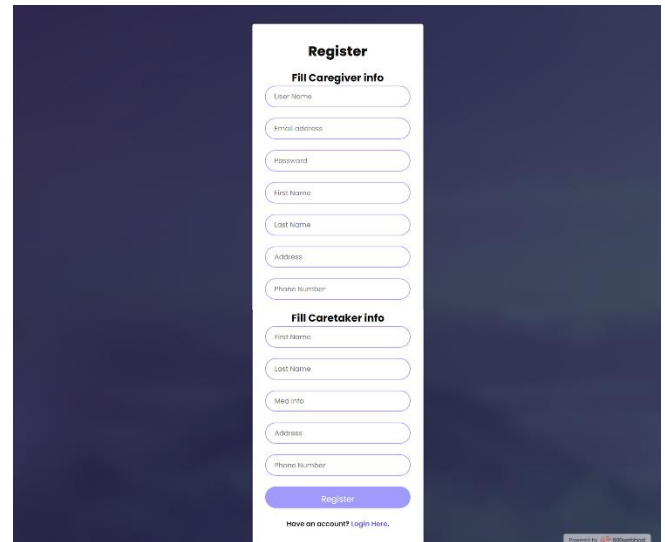


Figure 4.3 Register page

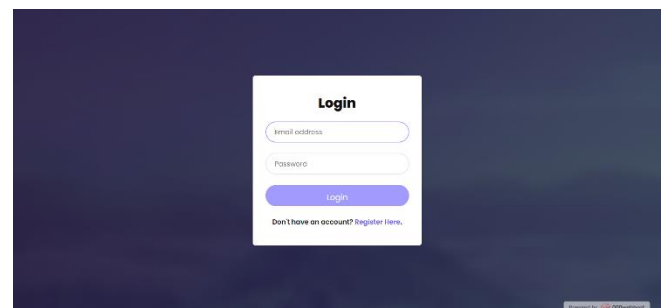


Figure 4.4 Login page

### 4.2.3 Index page

Once logged in, you will be able to see an index page, in which this index page will display menu bars, About Info sub-menu and log out, as shown in Figure 4.5

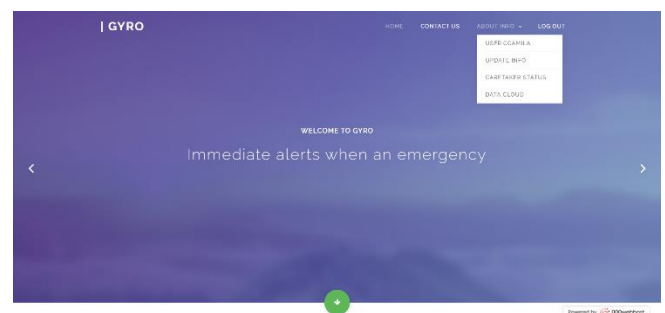


Figure 4.5 Index Page

#### 4.2.3.1 About Info sub-menu

##### a) Update info page

By clicking About Info sub-menu and then clicking Update Info, it will lead to an updated information page which can see all the information that can be updated the information of caregiver and caretaker as shown in figure 4.6.

**Update Info**

**| Caregiver info**

Username:

Email:

First Name:

Last Name:

Address:

Address:

Phone Number:

**| Caretaker info**

First Name:

Last Name:

Med info:

Address:

Phone Number:

[Back Home](#)

Figure 4.6 Update Info Page

##### b) Caretaker status page

By clicking About Info sub-menu and then clicking Caretaker status, it will show about address and status of caretaker in a graph as shown in figure 4.7.

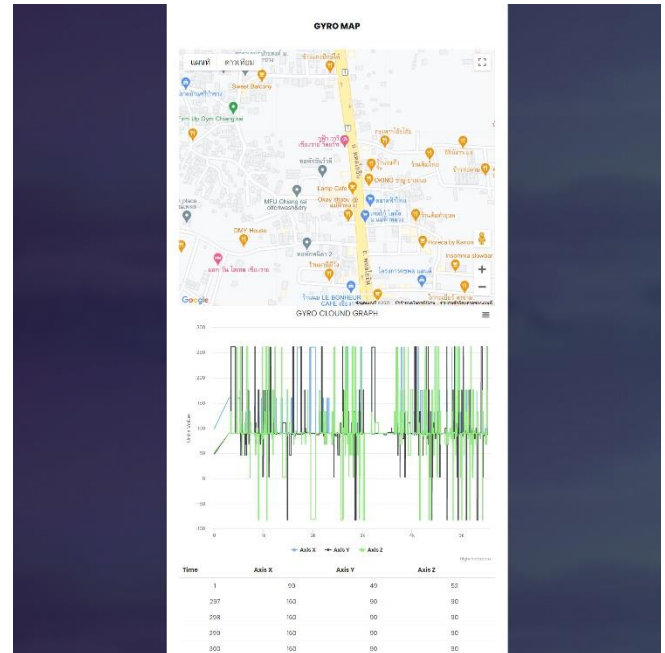


Figure 4.7 Caretaker status page

##### c) Data cloud page

By clicking About Info sub-menu and then clicking Data cloud, it will show values of gyroscope sensor, the status of caretaker, latitude and longitude, location name, and real-time timestamp in the table as shown in figure 4.8.

**GYRO CLOUD**

| Time | Axis X | Axis Y | Axis Z | Status | Latitude | Longitude | LOCATION NAME | Timestamp           |
|------|--------|--------|--------|--------|----------|-----------|---------------|---------------------|
| 9532 | 91     | 88     | 90     | 0      | 0        | 0         | Home          | 2020-05-07 22:03:04 |
| 9535 | 91     | 88     | 90     | 0      | 0        | 0         | Home          | 2020-05-07 22:03:53 |
| 9536 | 91     | 88     | 90     | 0      | 0        | 0         | Home          | 2020-05-07 22:04:43 |
| 9534 | 91     | 88     | 90     | 0      | 0        | 0         | Home          | 2020-05-07 22:03:33 |
| 9533 | 90     | 89     | 90     | 0      | 0        | 0         | Home          | 2020-05-07 22:03:23 |
| 9532 | 90     | 89     | 90     | 0      | 0        | 0         | Home          | 2020-05-07 22:02:13 |
| 9531 | 90     | 89     | 90     | 0      | 0        | 0         | Home          | 2020-05-07 22:02:03 |
| 9530 | 90     | 89     | 90     | 0      | 0        | 0         | Home          | 2020-05-07 22:01:53 |
| 9529 | 91     | 88     | 90     | 0      | 0        | 0         | Home          | 2020-05-07 22:01:10 |

Figure 4.8 Data cloud page

### 4.3 Line Notification

#### 4.3.1 Registration of caregiver of Line Application

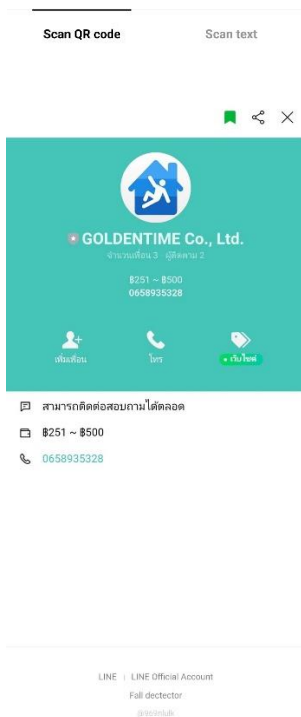
In the first use of a fall detector, cg is required to fill out the CT details, including the details of who would like the device to



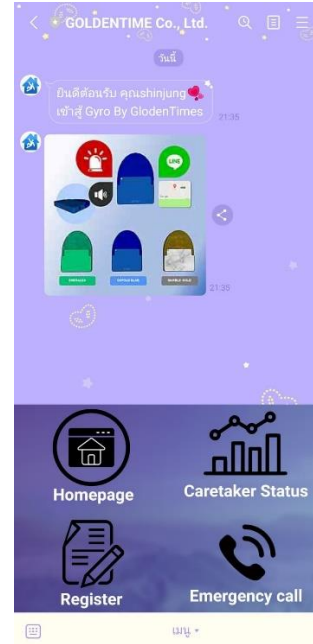
send a warning message if a fall occurs or to seek assistance from the operator of the device. User registration can be done by opening Line Application and adding friends by scanning the QR code as shown in figure 4.9 (a). After that, a new window will appear and press the button "Add friend" as in figure 4.9 (b). Then a window will appear showing a Greeting message window in figure 4.9 (c).



Just scan a QR code for a variety of LINE features, such as adding friends or using Rabbit LINE Pay.



(a)



(c)

Figure 4.9 Steps to add GOLDENTIME Co., Ltd. friend

(a) Scanning QR Code to add friends

(b) GOLDENTIME Co., Ltd. friend page

(c) Greeting message window

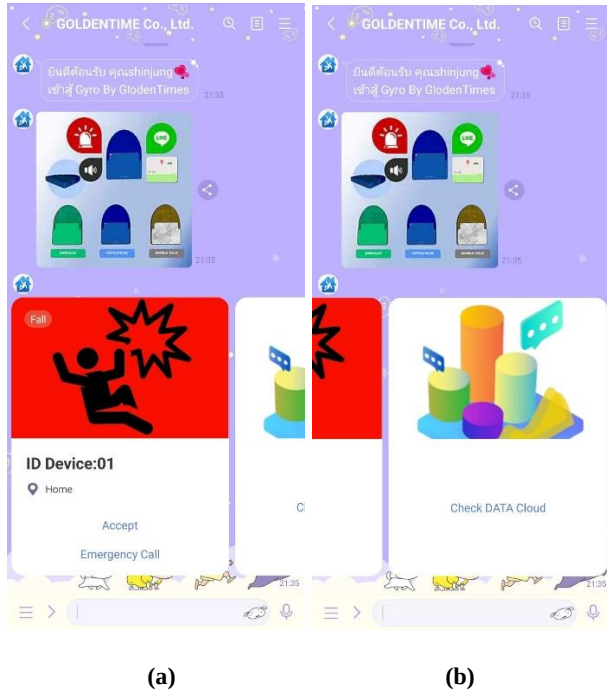
When clicking the Register button, a web browser window will appear for the caregiver to fill in the information of caregiver including first name, last name, email, address and telephone number and fill in the information of caretaker including first name, last name, medical information, address and telephone number as shown in figure 4.10.

Figure 4.10 Register page



#### 4.3.2 Notification messages within the Line Application

Notifications via Line Application in the form of pictures, text. And when there is an alarm in the event of a fall, there will be a button to ask for help, the notification message will be in the form of Button template as shown in figure 4.11(a) and can check caretaker real-time status by clicking on the 'Check DATA Cloud' as shown in figure 4.11(b).



**Figure 4.11** Notification messages within the Line Application

### 4.4 Experiment

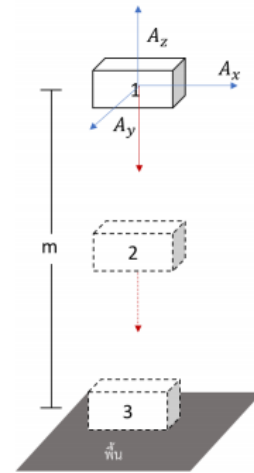
#### 4.4.1 Demonstrate Percentage

This project conducted a free fall experiment to see the behavior of the accelerometer sensor (MPU6050) shown in Figure 4.12 shows the free fall of the fall detector.

Position1 is the time when the device is at rest above the ground. After that, when starting to release the device, which will put the device in a free fall position2 This is the period when the device is in a free fall situation.[6]

Position3 is when the device hits the ground. Considering graph 4.12, it can be noted that if the device is in position1, the total acceleration is approximately 1g or 9.8 m/s<sup>2</sup> is the acceleration that acts on the device when the device is released, the acceleration value will decrease until it approaches 0 m/s<sup>2</sup>. Is that the device is on the low Position2 and when the device has landed in the3 the acceleration will vary as the device hits the ground. If the device falls and freezes it will revert to its initial state, with the acceleration acting on the device 9.8 m/s<sup>2</sup>.

$$A_{sum} = \sqrt{A_x^2 + A_y^2 + A_z^2}$$



**Figure 4.12** Shows the free fall of the fall detection device

The acceleration in the x, y and z axis, 3-axis acceleration sensor is derived from the calculation of the equation, and can be analyzed at all. The 4-port test failure was falling forward. Falling backwards, falling to the left and to the right, fall ten times repeated the experiment done during the fall Table 4.1

**Table 4.1** Table of acceleration while falling

| Fall | Fall forward | Fall back | Fall left | Fall right |
|------|--------------|-----------|-----------|------------|
| Time |              |           |           |            |
| 1    | 2.31         | 1.37      | 2.24      | 1.76       |
| 2    | 2.27         | 1.44      | 2.43      | 1.57       |
| 3    | 2.24         | 1.56      | 2.56      | 1.29       |
| 4    | 2.37         | 1.36      | 2.43      | 1.88       |
| 5    | 2.00         | 1.51      | 1.35      | 1.78       |
| 6    | 2.45         | 1.36      | 2.24      | 1.79       |
| 7    | 2.63         | 1.24      | 2.43      | 1.89       |
| 8    | 2.31         | 1.36      | 1.07      | 1.88       |
| 9    | 2.51         | 1.36      | 2.12      | 1.81       |
| 10   | 2.41         | 1.36      | 1.49      | 1.86       |
| Mean | 2.35         | 1.39      | 2.04      | 1.76       |

### 5. CONCLUSION AND FUTURE WORK

In this project, we have created a system example. Set the sensor Used in measuring the value of Ee in order to calculate the user's fall through control using Arduino Programming to control the work data collected in the Database for accuracy were easy to readed.

We have connected the database system and the operation instructions from the device and our website through the Line Application to allow the user to use and connect to the device easily.

For future work, We will develop a working system starting from developing the sensor and the shape of the device to be efficient in use and the shape of the device to be interesting in use. Including a fall detection system that is tough and speed of alarm response faster and more agile. In response to the use and continue to expand in the business and actually sell in the market.

#### REFERENCES

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